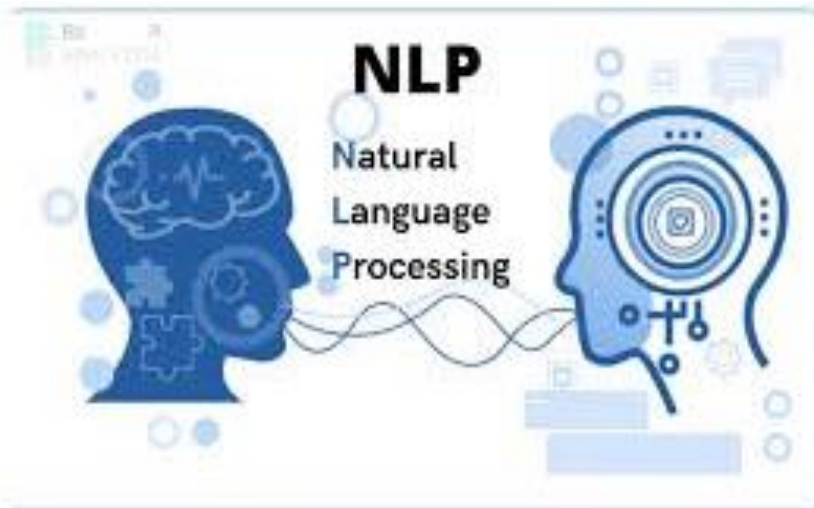




Natural Language Processing

Lecture 3:

Regular Expressions and Finite State Automata



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EXERCISE

1. Remove all digits from the text.

“I scored 95 marks in NLP 2025”

2. Extract all hashtags from the text.

“I love #NLP and #MachineLearning”

3. Tokenize the sentence using regex (ignore punctuation).

“Regex-based NLP is powerful!”

4. Remove all punctuation from the sentence.

"Hello!!! Welcome to NLP, 2026."

5. Extract all numbers from the text.

"Batch 2023 has 45 students and 3 instructors"

6. Normalize spaces to a single space.

"This is an NLP class"

7. Extract User Mentions

"Thanks @alice and @bob123"

8. Extract dates from text.

"The exam is on 12-03-2026"

9. Split text into sentences.

"NLP is fun. Regex is powerful! Let's learn."

10. Build a regex-based preprocessing Python function that:

- 1. Removes URLs**
- 2. Removes punctuation**
- 3. Tokenizes text**

```
print(preprocess("Learn NLP at https://nlp.ai! It's awesome."))
```

Mini Project Exercise

Regular Expressions for NLP with Python

Instructions

- Use Python and the **re** module only.
- Write clean, well-commented code.
- Do not use NLP libraries (NLTK, spaCy, etc.).
- Assume the input text is given.

Hello!!! I am studying NLP in 2026 😊.

Contact me at student123@gmail.com or admin@ucspyay.edu.mm.

Visit <https://nlp-course.org> for details.

The exam will be held on 12-03-2026.

NLP is soooo coool!!!

Questions

1. Using `re.search()`, write a Python statement to check whether the given text contains any URL. Print "URL found" if a URL exists.
2. Using `re.match()`, check whether the given text starts with the word "Hello". Print an appropriate message.
3. Using `re.findall()`, extract all email addresses from the given text and store them in a list.
4. Using `re.finditer()`, extract the date in DD-MM-YYYY format and print:
 - the date
 - starting index
 - ending index

Questions

5. Using `re.sub()`, remove all URLs from the given text.
6. Using regex:
 - Remove emojis
 - Remove punctuation
 - Remove digits
 - Display the cleaned text.

Questions

7. Using `re.sub()`, normalize repeated characters so that no character appears more than two times consecutively.
8. Using `re.split()`, split the text into individual sentences.
9. Using `re.compile()`, tokenize the cleaned text into words and convert them to lowercase.

Questions

10. Write a Python **function `regex_nlp_pipeline(text)` that performs the following steps in order:**

- **Remove URLs**
- **Remove punctuation and emojis**
- **Remove numbers**
- **Normalize repeated characters**
- **Tokenize the text**

Return the final list of tokens.